# MATH120-24S2 Discrete Mathematics

## Tutorial 8

### References

- Topics 13-14: Functions and Cardinality
- Sections, 7.1 – 7.4 of Epp (4th or 5th Ed.)

Work on the following problems in groups during the tutorial in week 9.

1. [D] Let $f : \mathbb{N} \to \mathbb{N}$ be the function given by the rule $f(n) = n^2 + 3$ and let $g : \mathbb{N} \to \mathbb{Z}$ be the function given by the rule $g(n) = 5n - 11$.

   (a) Calculate $f(2)$, $f(3)$, $f(5)$, $g(2)$, $g(3)$, and $g(5)$.

   (b) For each of the functions $f$ and $g$, what is the domain, co-domain, and range?

   *Solution:*

   (a) $f(2) = 7$, $f(3) = 12$, $f(5) = 28$, $g(2) = -1$, $g(3) = 4$, $g(5) = 14$

   (b) Domain of $f$ is $\mathbb{N}$, co-domain of $f$ is $\mathbb{N}$, and range of $f$ is

   $$\{3, 4, 7, 12, \ldots\}.$$

   Domain of $g$ is $\mathbb{N}$, co-domain of $g$ is $\mathbb{Z}$, and range of $g$ is

   $$\{-11, -6, -1, 4, 9, 14, \ldots\}.$$

$\Longrightarrow$ 2. [B-M] The function $f(m, n) = 2^m 3^n$ is a one-to-one function from $\mathbb{N} \times \mathbb{N}$ into $\mathbb{N}$.

   (a) Calculate $f(m, n)$ for five different elements $(m, n)$ of $\mathbb{N} \times \mathbb{N}$.

   (b) Explain why $f$ is one-to-one.

   (c) Is $f$ onto?

   (d) Show that $g : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ defined by $g(m, n) = 2^m 4^n$ is not one-to-one.

*Solution:*

(a) $f(0,0) = 2^0 \cdot 3^0 = 1$, $f(0,1) = 2^0 \cdot 3^1 = 3$, $f(1,1) = 2^1 \cdot 3^1 = 6$, $f(1,2) = 2^1 \cdot 3^2 = 18$, $f(2,2) = 2^2 \cdot 3^2 = 36$

(b) Suppose that $f(m_1, n_1) = f(m_2, n_2)$. Then

$$2^{m_1} \cdot 3^{n_1} = 2^{m_2} \cdot 3^{n_2}.$$

By the Fundamental Theorem of Arithmetic, every integer $n \geq 2$ can be written as a unique product of prime numbers. Thus, as

$$2^{m_1} \cdot 3^{n_1} = 2^{m_2} \cdot 3^{n_2}$$

and each side of the equality is written as a product of primes, the number of $2$'s on the left-hand-side of the equality is the same as the number of $2$'s on the right-hand-side of the equality. Similarly, the number of $3$'s also equate. Hence $m_1 = m_2$ and $n_1 = n_2$, and so $f$ is one-to-one. How cute is that!

(c) $f$ is not onto as $5 \neq 2^m \cdot 3^n$ for any choice of $m, n \in \mathbb{N}$. Why?

(d) $g(3,0) = 2^3 \cdot 4^0 = 8$ and $g(1,1) = 2^1 \cdot 4^1 = 8$, but $(3,0) \neq (1,1)$. Thus $g$ is not one-to-one.

$\Longrightarrow$ 3. [B] Let $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 3x + 1$. Prove that $f$ is a bijection and give its inverse.

*Solution:*

We first show that $f$ is one-to-one and onto.

Suppose $f(a) = f(b)$. Then $3a + 1 = 3b + 1$, which implies $3a = 3b$ and hence $a = b$ (by cancellation property). Hence $f$ is one-to-one.

Consider $y \in \mathbb{R}$. We note that $(y-1)/3 \in \mathbb{R}$ for all $y \in \mathbb{R}$. Then $f((y-1)/3) = 3((y-1)/3) + 1 = y$ for all $y \in \mathbb{R}$ and hence $f$ is onto.

Therefore $f$ is a bijection and $f^{-1}(x) = (x-1)/3$.

4. [B] Let $A$ and $B$ be finite sets, and suppose $|A| < |B|$. For each of the following statements, decide whether it is true or false.

(a) There is a map from $A$ to $B$ that is one-to-one.

(b) There is a map from $B$ to $A$ that is one-to-one.

(c) There is a map from $A$ to $B$ that is onto.

(d) There is a map from $B$ to $A$ that is onto.

*Solution:*

(a) True. List the elements of $A$ and list the elements of $B$. Map the first element of $A$ to the first element of $B$, map the second element of $A$ to the second element of $B$ and so on until every element of $A$ is mapped to an element of $B$. We can do this because $|A| < |B|$. The resulting mapping (function) is one-to-one.

(b) False. If there is such a mapping $f$, then, for each element in $A$, there is a distinct element in $B$ mapped to it via $f$. But then $|A| \geq |B|$; a contradiction.

(c) False. If there is such a mapping $g$, then, for each element in $B$, there is an element in $A$ mapped to it. Since $g$ is a map (function), each element in $A$ maps to precisely one element in $B$. Thus $A$ has at least many members as $B$; a contradiction.

(d) True. As in (i), list the elements of $A$ and list the elements of $B$. Map the first element of $B$ to the first element of $A$, the second element of $B$ to the second element of $A$, and so on until each element of $A$ is the image of an element of $B$. Now map the remaining elements of $B$ to any element of $A$.

$\Longrightarrow$ 5. [B-M] Let $5\mathbb{Z}$ denote the set of elements in $\mathbb{Z}$ divisible by $5$. Thus

$$5\mathbb{Z} = \{\ldots, -10, -5, 0, 5, 10, \ldots\}.$$

(a) Show that $\mathbb{Z}$ and $5\mathbb{Z}$ have equal cardinality.

(b) Using (a), prove that $5\mathbb{Z}$ is countable.

*Solution:*

(a) Let $f : \mathbb{Z} \to 5\mathbb{Z}$ be the function defined by setting $f(n) = 5n$ for all $n \in \mathbb{Z}$. We will show that $f$ is a bijection.

To see that $f$ is onto, let $m$ be an element of $5\mathbb{Z}$. Then $m = 5k$

for some $k \in \mathbb{Z}$. In particular, $f(k) = 5k = m$. Therefore $f$ is onto.

To show that $f$ is one-to-one, suppose that $f(n_1) = f(n_2)$. Then $5n_1 = 5n_2$ and so $n_1 = n_2$. Hence $f$ is one-to-one. It now follows that $f$ is a bijection and so $5\mathbb{Z}$ and $\mathbb{Z}$ have equal cardinality.

(b) By Example 14.6 in the lecture notes, $\mathbb{Z}$ is countable, that is, $\mathbb{Z}$ has equal cardinality with $\mathbb{Z}^+$. Since equal cardinality is an equivalence relation, we deduce by transitivity that $5\mathbb{Z}$ has equal cardinality with $\mathbb{Z}^+$. In other words, $5\mathbb{Z}$ is countable.

6. [M] Determine whether the following functions are one-to-one or onto. Prove your answer or give a counterexample.

(a) $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = x^3$.

$\Longrightarrow$ (b) $g : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ defined by $g(m, n) = 2m + 3n$.

(c) $h : \mathbb{R} \to \mathbb{R}$ defined by $h(x) = x|x|$. (Recall that $|x|$ is the absolute value of $x$, e.g. $|2| = 2$ and $|-2| = 2$.)

$\Longrightarrow$ (d) $k : \mathbb{R} \to \mathbb{R}$ given by $k(x) = 2^{x-1} + 3$.

*Solution:*

(a) $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = x^3$.
$f$ is one-to-one. Suppose $f(a) = f(b)$, then $a^3 = b^3$ and hence $a = b$.
$f$ is onto. Note that $\sqrt[3]{y} \in \mathbb{R}$ for all $y \in \mathbb{R}$ (note that $\sqrt[3]{-y} = -\sqrt[3]{y}$). Therefore, $f(\sqrt[3]{y}) = y$ for all $y \in \mathbb{R}$.

(b) $g : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ defined by $g(m, n) = 2m + 3n$.
$g$ is not one-to-one. For example $g(4, 1) = 11 = g(2, 3)$ but $(4, 1) \neq (2, 3)$.
$g$ is not onto. There is no element $(m, n) \in \mathbb{N} \times \mathbb{N}$ such that $2m + 3n = 1$.

(c) $h : \mathbb{R} \to \mathbb{R}$ defined by $h(x) = x|x|$.
$h$ is one-to-one. Suppose $h(x) = h(y)$, then $x|x| = y|y|$. It follows that either $x$ and $y$ are both positive or both nonnegative. (To see this, note that if $x < 0$ and $y > 0$, then $x|x| = -x^2 < 0$ and

$y|y| = y^2 > 0$ which contradicts $h(x) = h(y)$.) If $x, y > 0$ then $x|x| = y|y|$ implies $x^2 = y^2$ and hence $x = y$. If $x, y \leq 0$ then $x|x| = y|y|$ implies $-x^2 = -y^2$ and hence $x = y$.

$h$ is onto. Let $y \in \mathbb{R}$, we show that there is some $x \in \mathbb{R}$ such that $h(x) = y$. If $y \geq 0$ and $y = x^2$, then $x = \sqrt{y} \in \mathbb{R}$ and $h(\sqrt{x}) = y$. If $y < 0$ and $y = x|x|$, then $x = -\sqrt{-y} \in \mathbb{R}$ and hence $h(-\sqrt{-y}) = (-\sqrt{-y})| - \sqrt{-y}| = -(-y) = y$.

(d) $k(x) : \mathbb{R} \to \mathbb{R}$ given by $k(x) = 2^{x-1} + 3$.
$k$ is one-to-one. Suppose $k(x) = k(y)$, then $2^{x-1} + 3 = 2^{y-1} + 3$ and it follows that $2^{x-1} = 2^{y-1}$. By taking $\log_2$ of both sides we obtain $x - 1 = y - 1$ and hence $x = y$. Therefore $k$ is one-to-one.

$k$ is not onto because there is no $x \in \mathbb{R}$ such that $2^{x-1} + 3 = 3$.

7. [M] Let $f : A \to B$ and $g : B \to C$ be functions.

$\Longrightarrow$ (a) Show that if $f$ and $g$ are both onto, then $g \circ f$ is onto.

(b) Show that if $f$ and $g$ are both one-to-one, then $g \circ f$ is one-to-one.

*Solution:*

(a) Suppose that $f$ and $g$ are onto. Let $c \in C$. Since $g$ is onto, there is an element $b \in B$ such that $g(b) = c$. Furthermore, as $f$ is onto, there is an element $a \in A$ such that $f(a) = b$. It now follows that

$$(g \circ f)(a) = g(f(a)) = g(b) = c,$$

and so $g \circ f$ is onto.

(b) Suppose that $f$ and $g$ are one-to-one. Suppose that $(g \circ f)(a_1) = (g \circ f)(a_2)$. Since $g(f(a_1) = g(f(a_2))$, we have that $f(a_1) = f(a_2)$ as $g$ is one-to-one. Furthermore, $f(a_1) = f(a_2)$ implies that $a_1 = a_2$ as $f$ is one-to-one. It now follows that $g \circ f$ is one-to-one.

$\Longrightarrow$ 8. [M] Use a diagonalisation argument to show that $\mathbb{N} \times \mathbb{N}$ is countable.

*Solution:*

We first arrange the elements of $\mathbb{N} \times \mathbb{N}$ in a lattice:

$$\begin{array}{ccccc}
(0,0) & (0,1) & (0,2) & (0,3) & \cdots \\
(1,0) & (1,1) & (1,2) & \cdots & \\
(2,0) & (2,1) & (2,2) & & \\
(3,0) & & & \ddots & \\
(4,0) & & & & \\
\vdots & & & &
\end{array}$$

Then we describe a function $f : \mathbb{Z}^+ \to \mathbb{N} \times \mathbb{N}$ given by $f(1) = (0,0), f(2) = (0,1), f(3) = (1,0), f(4) = (2,0), f(5) = (1,1), f(6) = (0,2), f(7) = (0,3), f(8) = (1,2), \ldots$

This list of elements in $\mathbb{N} \times \mathbb{N}$ hits each element exactly once, that is, if $a \neq b$ then $f(a) \neq f(b)$ and for each $(m,n) \in \mathbb{N} \times \mathbb{N}$ there is some $a \in \mathbb{Z}^+$ such that $f(a) = (m,n)$. Therefore $f$ is a bijection.

Since $f : \mathbb{Z}^+ \to \mathbb{N} \times \mathbb{N}$ is a bijection, the sets $\mathbb{Z}^+$ and $\mathbb{N} \times \mathbb{N}$ have the same cardinality so $\mathbb{N} \times \mathbb{N}$ is countably infinite.

*Can you construct another proof that doesn't use the diagonalisation argument?*

9. [M-C] Let $S = \{x \in \mathbb{R} : \ 0 < x < 1\}$.

   (a) Let $V = \{x \in \mathbb{R} : \ -5 < x < -3\}$. Prove that $S$ and $V$ have the same cardinality.

   (b) Let $a$ and $b$ be real numbers with $a < b$, and suppose that

   $$W = \{x \in \mathbb{R} : \ a < x < b\}.$$

   Prove that $V$ and $W$ have the same cardinality.

*Solution:*

(a) Let $f : S \to V$ be the function defined by setting

$$f(x) = -2x - 3.$$

We first show that $f$ is a bijection.

To see that $f$ is onto, let $z \in V$. Now $x = -\frac{z+3}{2} \in S$ and

$$f(x) = f(-\frac{z+3}{2}) = -2.(-\frac{z+3}{2}) - 3 = z + 3 - 3 = z,$$

so $f$ is onto.

For showing that $f$ is one-to-one, suppose that $f(x_1) = f(x_2)$. Then

$$-2x_1 - 3 = -2x_2 - 3,$$

and so $x_1 = x_2$. So $f$ is one-to-one. Thus $f$ is a bijection, and so $S$ and $V$ have the same cardinality.

(b) Similar to (a), but choose $f$ to be the function from $W$ to $S$ defined by setting

$$f(x) = \frac{x-a}{b-a}.$$

This shows that $W$ and $S$ have equal cardinality. As $S$ and $V$ have equal cardinality and having equal cardinality is an equivalence relation, $V$ and $W$ have equal cardinality.

10. [C] Let $\mathbb{N}$ be the set $\{0, 1, 2, \ldots\}$ of natural numbers and let $\mathcal{P}(\mathbb{N})$ be the power set of $\mathbb{N}$, that is, $\mathcal{P}(\mathbb{N})$ is the collection of all subsets of $\mathbb{N}$. Show that there is no bijection between $\mathbb{N}$ and $\mathcal{P}(\mathbb{N})$. (This shows that, although $\mathbb{N}$ is countable, $\mathcal{P}(\mathbb{N})$ is uncountable.)

*Hint.* Suppose $f : \mathbb{N} \to \mathcal{P}(\mathbb{N})$ is a bijection. Let

$$A = \{x \in \mathbb{N} : \; x \notin f(x)\}.$$

Then $A$ is a subset of $\mathbb{N}$, so $A \in \mathcal{P}(\mathbb{N})$. Since $f$ is onto, there is a $z \in \mathbb{N}$ such that $f(z) = A$. Now obtain a contradiction.

*Solution:*

Suppose that $f : \mathbb{N} \to \mathcal{P}(\mathbb{N})$ is a bijection. Let

$$A = \{x \in \mathbb{N} : \; x \notin f(x)\}.$$

Then $A$ is a subset of $\mathbb{N}$, so $A \in \mathcal{P}(\mathbb{N})$. Since $f$ is onto, there is a $z \in \mathbb{N}$ such that $f(z) = A$.

We will derive a contradiction to the existence of $z$ (and hence, to the surjectivity of $f$) by showing that $z$ should at the same time be in $A$ and not in $A$.

If $z \in A$, then the definition of $A$ implies that $z \notin f(z)$. But $f(z)$ is equal to $A$, so if $z \notin f(z)$, we have that $z \notin A$. We have deduced from $z \in A$ that $z \notin A$, a contradiction (likewise, if $z \notin A$, then $z \in f(z)$ and $f(z) = A$, so $z \in A$). This shows that the existence of the element $z$ is contradictory, showing that $f$ is not a bijection.

*We can see that this argument does not use any properties of $\mathbb{N}$. So the same argument can be used to prove that the power set of a set $S$ cannot have equal cardinality with the set $S$ itself. Repeating this idea by taking $S$, $\mathcal{P}(S)$, $\mathcal{P}(\mathcal{P}(S))$, $\mathcal{P}(\mathcal{P}(\mathcal{P}(S)))$ etc... gives us infinitely many sets with different cardinalities! Can we go even 'larger', that is, are there sets that have cardinality different to that of some power set created by our process? The answer is yes! (Think about the union of all the sets we have already created.) This is all part of a very rich theory–Google 'cardinal numbers and power sets' to find out more.*